



LAUNCH READINESS / EVIDENCE REPORT

Dueling Dice

Windows launch QA

Prepared for the Monday, 28 September launch.

All five base modes and the tutorial completed in source play.
Build 340 results: pages 11-16. Test conclusions and ship decision: pages 17-22.

**5 / 5**

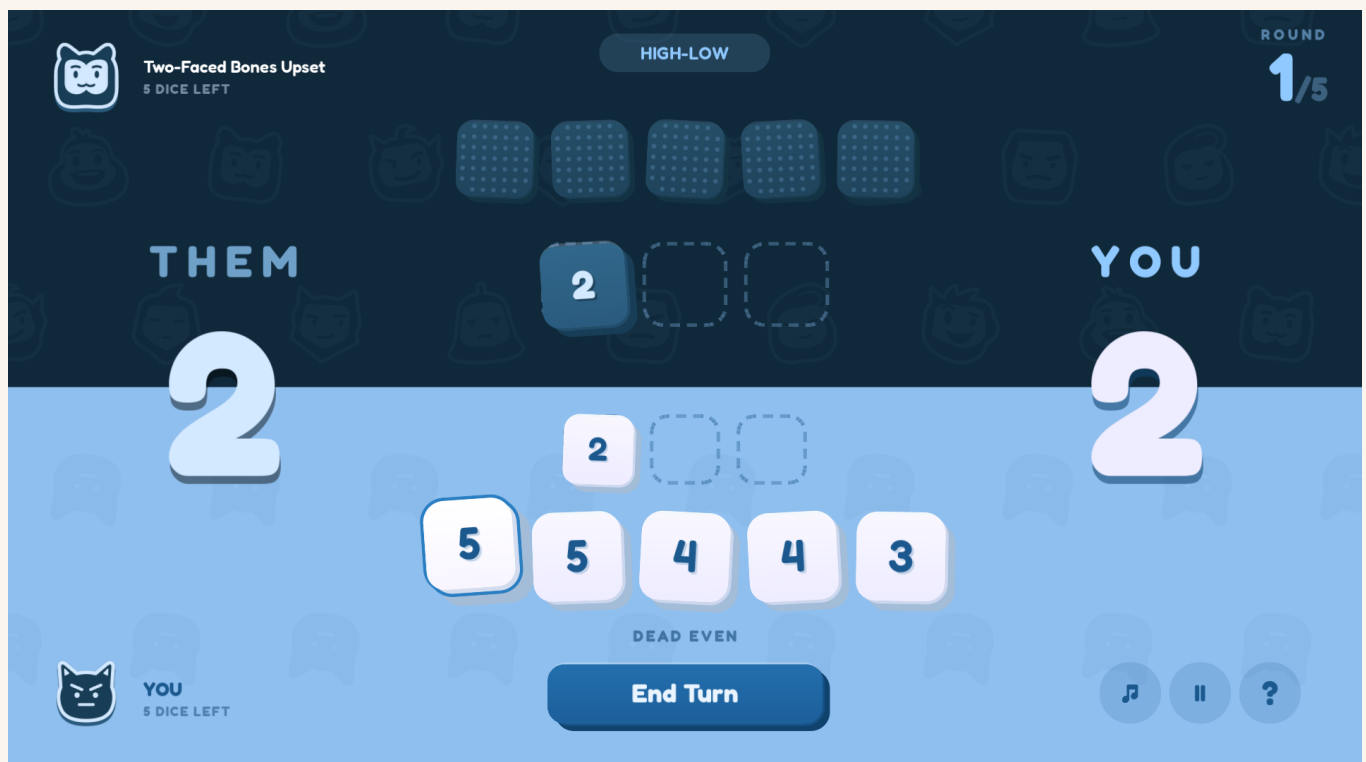
BASE MODES COMPLETED

840

ASSERTIONS / ENGINE

34,700

INITIAL FRAME SAMPLES



Captured during the MCP playtest: High-Low, 1920 x 1080. Source/debug build.

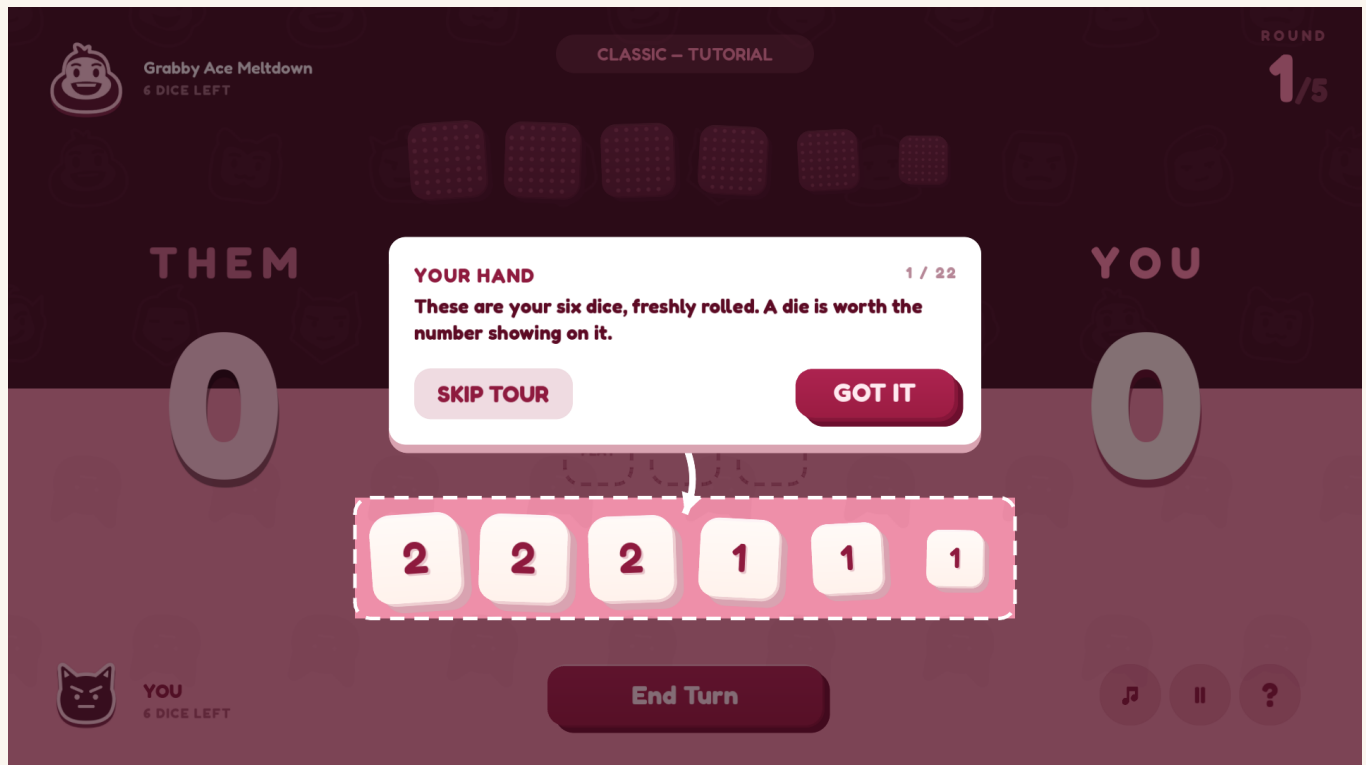
Ship-readiness recommendation: HOLD

Release candidate; final sign-off conditions are on pages 21-22.

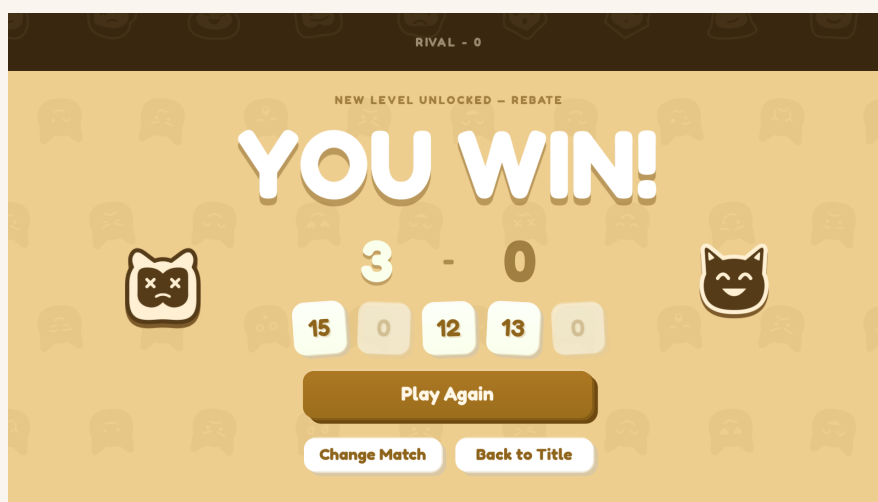
01 / PLAYTEST EVIDENCE

From the first hand to game over

Mouse-driven turns, a complete tutorial, and real match outcomes.



01 / Tutorial. The 22-beat story completed; its practice state was discarded before the real match reset to round 1 at 0-0.



02 / Tokens result. Screenshot from the source/debug session.

A win and an unlock

Tokens reached a win screen with Rebate unlocked. Both wins and losses were observed across the five modes.

Mode coverage: Classic, Sign Duel, Tokens, Mixed Hand and High-Low all reached game over. Later modes were selected through Session state to bypass unlock grinding; actual turns used MCP mouse input. This does not prove every progression unlock through ordinary play.



02 / P2 PERFORMANCE FINDING

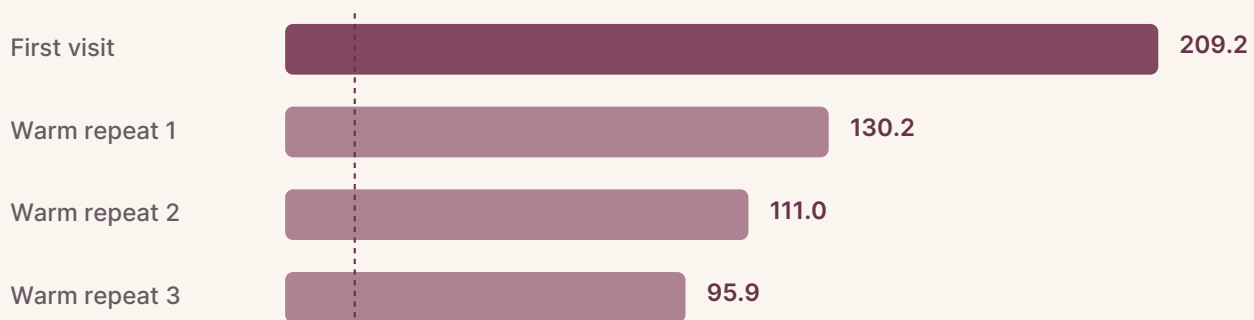
The menu transition needs attention

Setup produced a long frame on the first visit and on three warm repeats.



03 / Set the Table. Repro: title > Play > setup Back > Play again.

PEAK FRAME TIME / MILLISECONDS



Dashed marker: 16.67 ms budget at 60 Hz. Warm repeats omitted screenshots.

Initial finding / follow-up now appended

These original figures came from Godot 4.8 with MCP and the recorder active. The new 4.7.2 release capture and controlled source profile are on pages 8-10.



03 / PERFORMANCE EVIDENCE

Normal play and measurement limits

34,700 frame samples across roughly ten minutes. Recording included pauses, inspection, scene transitions and capture operations. Regression tests ran after recording ended.

Phase	Median	p99	Maximum
Sign Duel	16.64 ms	29.50 ms	94.57 ms
Tokens	16.58 ms	28.16 ms	91.43 ms
Mixed Hand	16.59 ms	29.29 ms	88.53 ms
High-Low	16.64 ms	30.89 ms	99.64 ms
Live board idle*	16.67 ms	20.66 ms	32.20 ms

Match phases include transition and capture overhead. *Idle sample: 25.4 seconds with no tool commands.

WHAT THE RECORDER MEASURED

Wall-clock frame intervals, CPU process time, static and video memory, draw calls, node counts and orphan counts. Live idle had no frame over 33 ms in the isolated sample.

INTERPRETATION

Screenshots, window resizing, eval/inspection and CSV flushing can add overhead. Their peaks should not be attributed automatically to gameplay. The warm-menu reproduction on page 3 deliberately excluded screenshot capture.

Static memory rose and later fell. The recorder itself retained samples, so this run does not establish a memory leak. Orphan counts returned to zero during stable idle; transient scene-change counts are not evidence of retained leaks.

Evidence retained

frames.csv and performance-summary.json contain the original samples and phase summaries. No long-duration soak was performed.



04 / ORIGINAL QA FINDINGS

Initial regression and tooling findings

Check	Result
Gameplay suite / Godot 4.8	836 assertions passed
Gameplay suite / Godot 4.7.2	836 assertions passed
Balance / balance rules	11 / 22 checks passed
Debug menu / idle guidance	14 / 95 checks passed
Spacing audit	20 elements inside the 72px margin; matches the existing count. Report-only.
Windows package startup	1,800 rendered frames; exit 0. No script errors.

P2 / THE REGRESSION GATE IS NOT ERROR-CLEAN

Both main-suite runs logged a missing **PlanHost**, detached-control **grab_focus** errors and exit cleanup diagnostics. The harness still returned success.

PlanHost: tools/test_modes.gd:2521 checks a removed node with `get_node()`. Use a non-erroring existence check.

Focus: determine whether deferred callbacks outlive test fixtures or live screens.

Cleanup: one CanvasItem RID, 23 ObjectDB instances and seven resources remained at exit.

Follow-up: correct test/scene teardown and make unexpected engine errors fail the release gate. Preserve the assertions. Normal MCP gameplay did not reproduce these errors.

UPSTREAM / MCP KEY INJECTION

Five `KEY_*` taps generated ten ClassDB lookup errors, although fallback key delivery worked. Plain Escape, Right and Enter produced no new errors and verified settings > pause > board, then a played-dice count of 0 > 1.

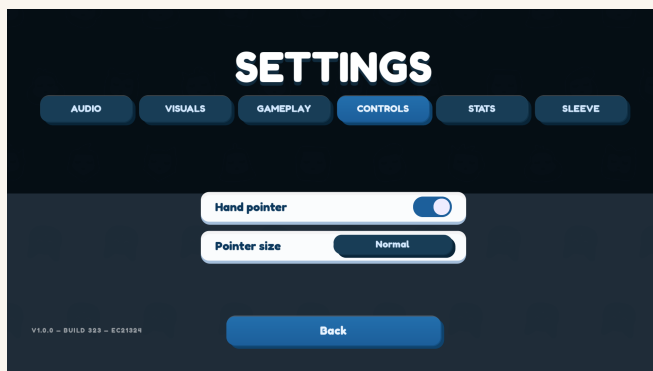
Originally recorded in upstream OPEN-ISSUES.md. The local fix and input-delivery checks are documented on page 11.



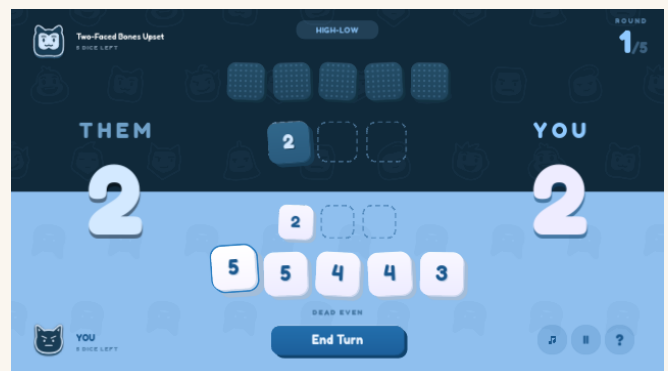
05 / VISUAL CHECKS

Desktop controls and layout

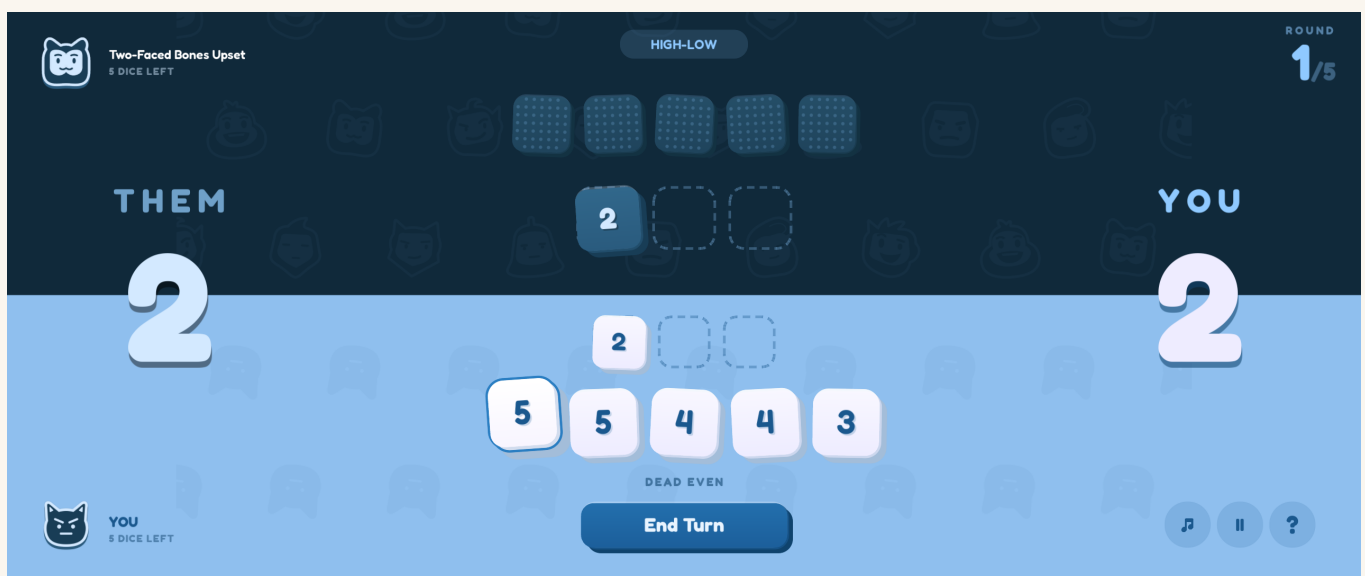
Pause and settings retained the match. Escape returned through settings and pause; Right/Enter selected and played a die.



04 / Controls settings. Audio and Visuals tabs were also checked.



05 / 640 x 360. Score, hand and controls remain in the viewport.



06 / Ultrawide, 2560 x 1080. The board retains its layout across the wider viewport.

Additional sizes inspected: 1280 x 720 and 1920 x 1080. No clipping was observed in the captured board views. Screenshots show the source/debug session, not the shipping executable.

Still needs hands-on verification

Physical controller behavior, fullscreen/Alt-Tab, multi-monitor behavior and audio by listening remain outside this pass.



06 / SCOPE AND HANDOFF

Build provenance and launch follow-up

Environment	Recorded configuration
Machine	Windows 11 Pro, build 26200 Ryzen 9 9955HX / RTX 4090 / about 94 GiB usable RAM
Interactive session	Godot 4.8.dev5.official.9552dfb68 D3D12 Forward+ / 1920 x 1080 / VSync on
Source commit	52b095c59c3550ecc7fab114aba43f63d1dca1e9 Displayed source stamp: build 323
Windows package	Version 1.0.0.338 Godot 4.7.2.rc.custom_build.36a04fe52

PACKAGED SMOKE TEST

The real GPU run loaded the title scene and exited normally after 1,800 frames. Two RGB8-to-RGBA8 conversion warnings were logged; no script error or crash accompanied them. A first launch from the source directory with `--path` exited before logging; launching from the package directory succeeded.

Original scope: the package excludes MCP. Full matches and initial frame telemetry came from the source/debug build; the new release profile is on pages 8-10. Separate APPDATA directories isolated saves. No game fixes, export-setting changes, commits or uploads were made.

BEFORE FINAL SIGN-OFF

Follow the profiling findings on pages 8-10 and play full matches in the final package. Test minimum-spec/integrated graphics, a physical gamepad, desktop window behavior, audio and a longer soak. Recheck the exact upload package if it is rebuilt.

EVIDENCE ARCHIVE

Source report, logs, captures and frame data:

`.godot/qa/2026-09-26/`

Logs retain original temporary paths. The archive was moved into the project after the run.

PACKAGE SHA-256

EXE 8ECD035601F5C71D5DD07C755358EC5EF4CE8402039EA5F11931D4A4E69F2FD2

PKC 12363A7BFB8E2AEF2F1DA6C4A7106FDE275DEC4C74EA257B207FF59EB3B08DE33



07 / FRAME-PROFILE ADDENDUM

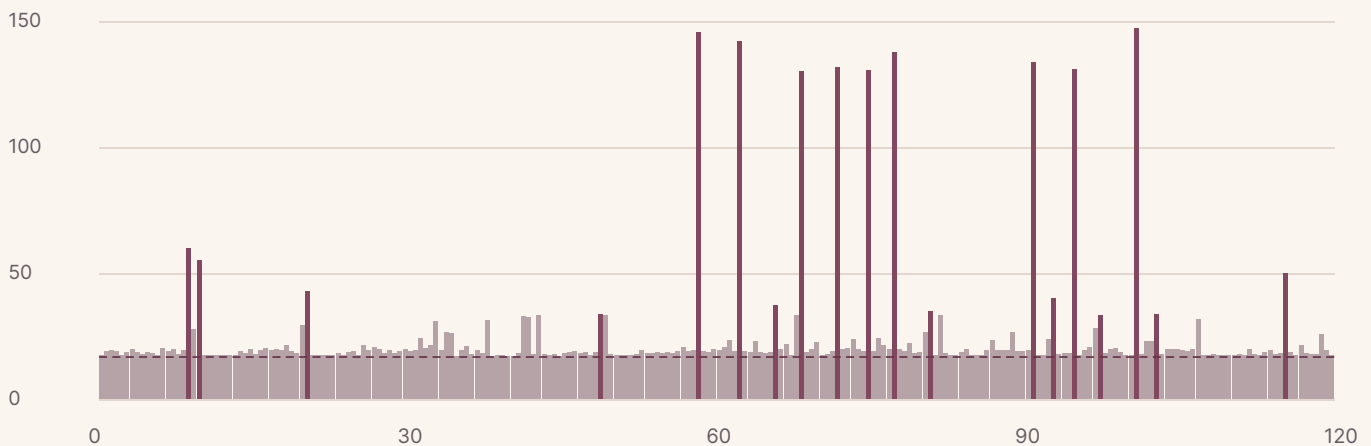
The release has measurable hitches

Build 338 / PresentMon 2.6.0 / 1920 x 1080 / VSync on. The two-minute user-operated capture contains 7,131 presents over 119.86 seconds.

Capture	Median	p99	Peak
Release / user-operated	16.67 ms	25.72 ms	147.20 ms
Release / startup	16.65 ms	20.86 ms	356.00 ms
Startup / after first 5 s	16.65 ms	19.50 ms	52.25 ms

User-operated capture: 24 intervals >33.33 ms, 11 >50 ms and 9 >100 ms.
Startup: 1,951 presents; tail: 1,705. Percentiles use nearest rank.

RELEASE CAPTURE / INTERVAL BETWEEN PRESENTS



Seconds since first captured frame. Each bar preserves the maximum in a 0.5 s bin.
Dashed line: 16.67 ms. Actions were not timestamped; peaks are not labeled as specific clicks.

Largest application interval: 147.20 ms. PresentMon recorded 132.99 ms CPU-side busy time and 35.12 ms GPU busy time for that row. Across all nine >100 ms intervals, CPU-side time was 114.95-132.99 ms versus 17.79-37.40 ms GPU busy.

A CPU-side stall is the stronger lead

These are elapsed timing metrics, not CPU utilization or sampled call stacks. The controlled source profile on page 9 identifies the expensive setup functions.



08 / CONTROLLED 4.7.2 SOURCE PROFILE

Setup rebuilds are the main lead

A rendered source harness called the real Play and Back handlers. Six visits used normal timing, no screenshot capture and no active script profiler. All visits produced a long frame.

Setup visit	Peak frame interval	Over 33.33 ms
First visit	139.69 ms	2
Warm 1	109.28 ms	1
Warm 2	138.44 ms	3
Warm 3	134.49 ms	1
Warm 4	138.10 ms	1
Warm 5	127.22 ms	1

SceneTree callback intervals, measured with a monotonic clock. The full timing run recorded 3,736 callbacks; PresentMon independently recorded 3,735 presents.

SEPARATE PROFILER RUN / SYNCHRONOUS FUNCTION TOTALS

Function	First	Warm 1	Warm 2
setup._ready (inclusive)	106.95 ms	101.22 ms	101.86 ms
_build_rival_row (inclusive)	76.16 ms	71.63 ms	73.25 ms
_hold_roster_art (self)	70.26 ms	66.44 ms	68.46 ms
_build_you (inclusive)	15.97 ms	15.98 ms	16.08 ms

The roster-loading function alone accounts for 65.6-67.2% of setup._ready. It made five calls per visit. The code synchronously loads texture paths into a dictionary owned by the setup scene; that cache is discarded when the screen is freed.

Profiler figures come from three separate 2 s windows. Inclusive rows overlap and must not be added. Async harness / transition totals include waits and are excluded from attribution. Profiling overhead means these are not release-frame timings.



09 / FINDINGS BEFORE FIXES

Fix the loading path, then remeasure

P2 / Repeatable setup construction hitch. The source profile isolates synchronous roster-art loading as the largest measured setup cost. Release telemetry independently confirms long intervals, but does not prove every release spike came from this function.

**WHERE TO INVESTIGATE**

scenes/setup/setup.gd:129 - `_ready` builds the screen.
scenes/setup/setup.gd:416 - `_build_rival_row`.
scenes/setup/setup.gd:457 - `_hold_roster_art`.
scenes/setup/setup.gd:95 - scene-owned `_roster_art` cache.
scenes/ui/mascot_head.gd:391 - texture-loading helper.

ORIGINAL RECOMMENDATIONS / IMPLEMENTED ON PAGE 11

Keep the roster texture set referenced across title/setup visits, or prewarm it before the transition. Measure the memory tradeoff and check that any asynchronous loading completes before scene construction. Reprofile player-head construction and wipe resources after the dominant load cost is reduced.

ACCEPTANCE RUN

Repeat ten Play > Back cycles in the rebuilt shipping package, then play each mode. Proposed target: no warm setup interval above 33.33 ms at 60 Hz on this machine. Retain raw traces and compare p99 plus worst-case timing; verify first-entry behavior separately and repeat on minimum-spec hardware.

MEASUREMENT BOUNDARIES

PresentMon captured the shipping executable without an injected recorder. Display-change intervals peaked at 216.68 ms; 104 rows lacked a display interval. Focus / compositor events were not marked, so gaps cannot all be assigned to game code. Native control was unavailable; manual action coverage was not independently verified.

The source runs used the 4.7.2 engine listed on page 7, fresh isolated saves and a test harness under `.godot`. Source content is not proven identical to build 338. No screenshot was taken during timed captures. Full-match completion, a long soak and low-spec performance remain unverified in the package.

Evidence: `.godot/qa/2026-09-26/frame-profile/` (raw CSVs, logs, harness, summary.json and script-attribution.json). Method: PresentMon console metrics documentation. Profiling left the code and package unchanged; subsequent fixes are recorded on page 11.



10 / FIX VERIFICATION - 26 SEPTEMBER

Source fixes applied and verified

Roster art now survives menu changes. Boot warms 50 texture paths incrementally and Session retains them. The screen uses the same assets and layouts.

Measurement	Before	After
Warm setup / peak range	109.28-138.44 ms	24.52-25.35 ms
First setup / peak	139.69 ms	49.29 ms
setup._ready / inclusive	101.22-106.95 ms	16.66-18.77 ms
Roster art / inclusive total	66.48-70.33 ms	0.066-0.077 ms

After: ten rendered visits on 4.7.2, through normal boot; nine warm visits, none above 33.33 ms. Before: six visits from the original source harness. Script totals are from separate profiled runs; these are source results, not a rebuilt shipping package.

FIXES AND CHECKS

Regression cleanup: guarded deferred focus, a non-erroring PlanHost existence check, explicit disposal of the test hint hand, and audio shutdown before quitting. Both Godot 4.7.2 and 4.8 passed **840 checks with no engine errors or shutdown leaks**. The new test gate rejects errors even after a PASS receipt.

MCP key input: fixed in the local upstream Swallowtail repo and backported to the game. Both engines passed 18 prefixed/plain names, press/release state, delivered events and nine taps with zero diagnostics, including keypad keys.

Memory tradeoff: clearing the retained menu cache after a separate profile reduced reported texture memory by **62.88 MiB** at the title. Boot warm-up completed in about 1.7 seconds on this machine.

Release retest follows

Build 340 was exported from the committed fixes; results follow on pages 12-16. Build 338 remains intact. Full packaged matches, minimum-spec hardware and cold-start spikes remain open.

Fix evidence: `.godot/qa/2026-09-26/fixes/` and gate logs under `.godot/qa/regression-*/`.



11 / RELEASE VERIFICATION - 26 SEPTEMBER

Build 340: export and package checks

The fixed source was committed as `cfcc0a04e23fea5eadb842a65fa06d8aaedc8828`. A Windows-only rehearsal exported the encrypted package, restored development settings, and staged the exact files to test. No store upload or tag was created.

Receipt	Result
Regression on export engine	840 checks; no engine errors or shutdown leaks
Encrypted package boot	Pass; 50 menu textures warmed in 1.15-1.66 s
Rendered package exits	Three clean exits; no shutdown diagnostics
New CLI rendered capture	1,799 presents; automatic checks pass; manual follow-up on page 15

COLD START / 35-SECOND EXTERNAL CAPTURE

Window	Presents	p99	Peak
All startup samples	1,930	20.99 ms	666.06 ms
After first five seconds	1,702	19.15 ms	48.96 ms

PresentMon 2.6.0; RTX 4090; D3D12 Forward+; 1920 x 1080; fresh isolated saves. The cold boot peak remains. This idle/startup capture does not verify repeated packaged menu transitions.

PACKAGE PROVENANCE

Desktop/Export/release/v1.0.0-build340/ship/windows/

EXE SHA-256: 487c73b9be7ac023fd1ecba003590f0a123b0f4e07ee12c45371dc273e0813cf

PCK SHA-256: b8e701c307d1018468022b02a2fd55cd9739edac675d5a85e5351470a74200bc

Open finding / headless shutdown diagnostics

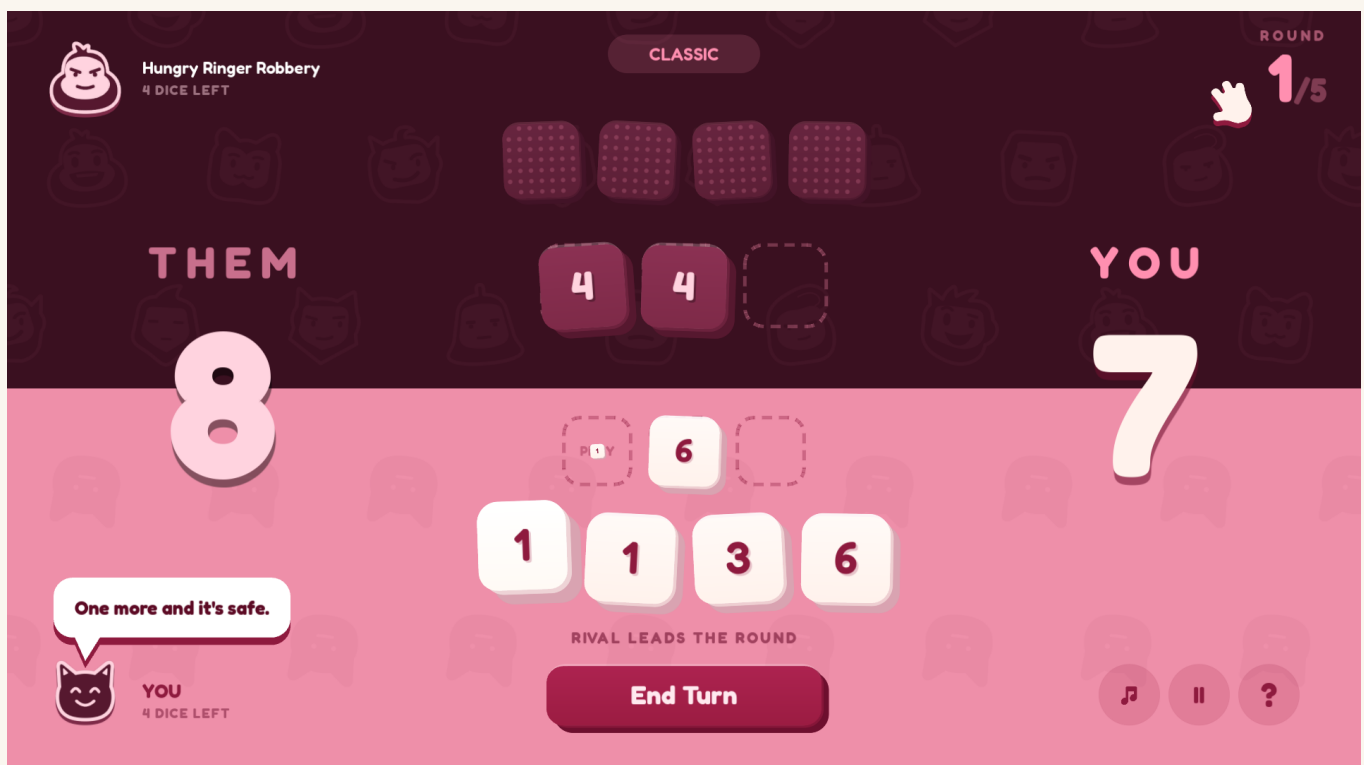
A repeated headless `--quit-after 600` run logged two leaked ObjectDB objects and one retained resource after exit 0. All three rendered package runs were clean. Retained as a failed receipt; scope is not yet established beyond headless teardown.

12 / SOURCE BASELINE AND SCREENSHOTS

The reusable scenario passes

46 checks passed. Normal boot, ten Play/Back cycles, two Classic die placements with rival replies, pause/resume and return to title. Screenshots ran in their own timing phase. These are source results; OS input delivery was not exercised.

Measurement	Latest source result
First setup / peak	49.49 ms
Nine warm setup visits / peaks	23.99-25.15 ms; none above 33.33 ms
Gameplay / median, p99, peak	16.62 / 18.55 / 19.81 ms
Complete process log	No engine errors or shutdown leaks



Classic match after two placements and rival replies. Captured by the new GDScript scenario; no screenshot work is included in the gameplay timing row.

Game-owned extension: `tools/qa_release.gd` and `tools/qa_release.json`. Swallowtail owns recording, log gates, hashing, comparison and PDF generation. The standalone scenario PDF was generated by the new built-in report command.



13 / REUSE IN OTHER GAMES

Swallowtail QA: first feature set

The first implementation lives in the upstream Swallowtail repository. Its local CLI embeds the runner, report template, fonts and mascot. Python 3.10+ runs the worker; ReportLab 4 renders the PDF.

Command	Purpose
qa init	Create a game-owned config and GDScript smoke scenario
qa run	Retain checks, logs, raw frames, screenshots and build hashes
qa compare	Compare compatible hardware, renderer and scenario contexts
qa attest	Record actual operator evidence for a declared manual check
qa report	Render the branded PDF from retained run artifacts

EXTENSIBLE WITHOUT GAME-SPECIFIC ASSUMPTIONS

A RefCounted script exposes `run(qa)`. Its SceneTree context provides `check`, `phase`, `wait` and `screenshot`. Editor custom commands can share game helpers while retaining their own registration model. Shipping package tests do not inject scripts.

VERIFIED AND DOCUMENTED

Go checks and parameter audit passed. Python contracts cover post-PASS errors, timeouts, missing captures, invalid metrics, modified evidence and incompatible baselines. The skill validator passed; the documentation site built 60 pages. The live source scenario and exact package runs exercise the feature against a real game.

**Remaining release coverage**

Page 15 records an unseen-window attempt; page 16 records the confirmed visible-window retest. Full packaged matches, physical controllers, long soak and minimum-spec hardware remain unverified. The native input helper was unavailable; source tests do not fill those gaps.



14 / MANUAL TEST NOT PERFORMED

Build 340: process capture only

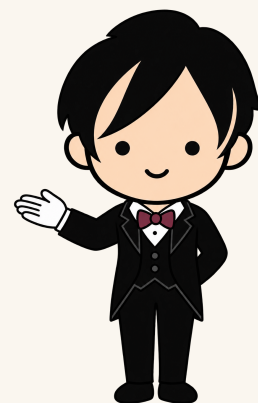
The exact package produced a 141.22-second presentation trace. The operator subsequently reported that the window never opened for them and they did not perform the test. This supersedes the earlier completion response. These metrics describe the process only; they do not establish gameplay coverage.

Capture window	Presents	p99	Peak
Complete trace	8,398	19.52 ms	665.09 ms
After first five seconds	8,169	19.43 ms	47.98 ms

After the first five seconds, **7 intervals exceeded 33.33 ms** and none exceeded 100 ms. All three intervals above 100 ms occurred in the initial five seconds. This is a whole-session summary, not per-action attribution.

AUTOMATED RECEIPTS

Process exit 0; no engine errors or shutdown leaks; boot receipt present; PresentMon exited cleanly. Four automated checks passed. EXE and PCK hashes match the build-340 provenance on page 12.



Manual test not performed

The operator reported no visible window. Both manual checks remain pending: five Play/Back cycles, and match turns with visual/audio review. Confirm a visible, interactive window before retrying. Release sign-off remains incomplete.

The native Windows control helper was unavailable, so no screenshots or input events were independently observed for this package session. PresentMon returned no usable click-to-photon rows. Menu-entry peaks cannot be identified from this trace alone. Source screenshots elsewhere in this report are explicitly labelled.



15 / VISIBLE WINDOW - OPERATOR EVIDENCE

Build 340: confirmed manual retest

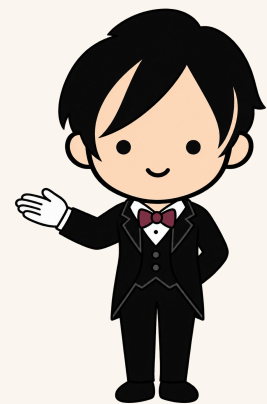
A fresh isolated build-340 process ran at requested 1280x720 without an automatic close timer. The operator confirmed the title screen was visible and interactive before a two-minute PresentMon capture attached to PID 10716. Boot is excluded from this trace.

Capture	Presents	Median	p99	Peak
Visible-window retest	7,175	16.66 ms	21.85 ms	126.03 ms

12 intervals exceeded 33.33 ms; 2 exceeded 100 ms. There are no per-action markers to attribute these spikes to menu entry, gameplay, focus or the compositor. The operator reported no noticeable issue; the measured spikes remain a finding.

CONFIRMED COVERAGE

The operator reported five Play/Back cycles, match turns, no freezing, missing art, input or audio issues noticed, and normal game closure. Four automated checks and three manual checks pass. Manual results are explicit operator attestations.



Clean shutdown; bounded checklist passed

Process exit 0. Full engine log: boot receipt present, no errors or shutdown leaks. Package hashes match page 12. These seven checks verify the declared coverage; no package-specific frame budget was configured.

This trace uses PresentMon v1 columns and 1280x720 after boot. It is not a like-for-like comparison with earlier 1920x1080 startup captures. Full matches, physical controllers, long soak and minimum-spec hardware remain unverified. Raw evidence and operator attestations: <output/qa/2026-09-26/package-visible/>.



Test conclusions and next actions

01 / Modes and tutorial

Conclusion: All five base modes and the 22-beat tutorial completed in source/debug play.

Fix status: No gameplay crash or softlock reproduced in that scope.

To finish: Complete a full match in each advertised mode and the tutorial in build 340, including restart and results.

Evidence: Launch report pp. 2, 5; initial playtest record.

02 / Progression and save continuity

Conclusion: Later modes were selected through Session state; ordinary unlock progression was not proven.

Fix status: No progression defect established; coverage remains incomplete.

To finish: Use the package to earn an unlock, exit, relaunch and verify unlocks, settings and progress persist.

Evidence: Launch report p. 2; initial playtest scope.

03 / Desktop controls and presentation

Conclusion: Source mouse/keyboard, pause/settings and sampled layouts passed. Package listening/visual review had no reported issue.

Fix status: No fix indicated for inspected views. Physical gamepad, fullscreen, Alt-Tab and multiple monitors remain unverified.

To finish: Exercise advertised controller support and desktop focus/window transitions; retain screenshots and input results.

Evidence: Launch report pp. 6, 16; package-visible/run.json.

04 / Repeated setup transition

Conclusion: The repeated source hitch is fixed in measured warm entries: nine peaks of 23.99-25.15 ms.

Fix status: Session retains 50 menu textures; boot warms them. Measured retention cost: 62.875 MiB.

To finish: Verify ten timestamped warm Play/Back cycles in the exact package against the proposed 33.33 ms peak budget; check memory on minimum hardware.

Evidence: Launch report pp. 9, 11, 13; navigation/frames.csv.



Test conclusions and next actions

05 / First setup entry

Conclusion: First source setup still peaked at 49.49 ms after the fix. Warm-entry success does not close this result.

Fix status: Improved from 139.69 ms; first-entry hitch remains.

To finish: Profile first entry separately; reduce remaining synchronous work or document an accepted first-entry budget and retest.

Evidence: Launch report pp. 9, 11, 13; navigation/run.json.

06 / Cold package startup

Conclusion: Build-340 startup traces peaked near 665-666 ms; after five seconds, the startup sample peaked at 48.96 ms.

Fix status: Not fixed or attributed to a specific function.

To finish: Check visible cold-start responsiveness on target hardware; localize the gap and fix it or explicitly accept a bounded loading-screen delay.

Evidence: Launch report pp. 12, 15; startup-presentmon.csv.

07 / Package frame consistency

Conclusion: The visible 1280x720 retest had p99 21.85 ms, peak 126.03 ms and two intervals above 100 ms.

Fix status: Unresolved timing finding; no issue reported by the operator. No per-action cause established.

To finish: Repeat with action/focus timestamps; investigate reproducible active-play stalls. Approve a package frame budget before declaring performance ready.

Evidence: Launch report p. 16; package-visible/presentmon.csv.

08 / Gameplay regression gate

Conclusion: 840 regression checks passed on both engines and the export gate. Initial balance (11), rules (22), debug-menu (14) and idle-guidance (95) checks also passed.

Fix status: Focus/PlanHost/test teardown defects fixed. Full-log gate rejects engine errors even after PASS.

To finish: Keep this gate mandatory on the final candidate; rerun after runtime changes. No further fix is indicated by these clean runs.

Evidence: Launch report pp. 5, 11; re-export.log; original and fixes/ gate evidence.



Test conclusions and next actions

09 / MCP key delivery

Conclusion: 18 named-key cases plus event/state checks and nine taps passed on both engines without new diagnostics.

Fix status: GlobalScope ClassDB lookup removed; named/keypad resolution fixed upstream and backported.

To finish: Retain the key regression check. This closes the tooling defect, not physical-controller coverage.

Evidence: Launch report p. 11; upstream input-key regression receipts.

10 / Reusable source scenario

Conclusion: All 46 checks passed: normal boot, ten menu visits, representative turns, pause/resume and return to title.

Fix status: No scenario failure remains. Gameplay p99 18.55 ms, peak 19.81 ms in its measured phase.

To finish: Retain this source baseline; extend full-match and save fixtures. Source method calls do not establish shipping input delivery.

Evidence: Launch report p. 13; navigation/run.json and screenshots.

11 / Export and rendered package lifecycle

Conclusion: Build 340 contains the fixes; rendered runs booted and closed cleanly. The visible retest exited 0 with no shutdown diagnostics.

Fix status: Re-export complete. Exact EXE/PCK hashes retained; no upload was made.

To finish: Freeze the tested package. Verify hashes of the staged upload and repeat applicable checks if rebuilt.

Evidence: Launch report pp. 12, 16; re-export.log; package-visible/engine.log.

12 / Headless package shutdown

Conclusion: A repeated headless package run logged two leaked ObjectDB objects and one retained resource despite exit 0.

Fix status: Open failure. Clean rendered exits do not explain or erase the headless result.

To finish: Reproduce and identify retained objects; fix and retest, or document evidence of headless-only scope and an explicit risk decision.

Evidence: Launch report p. 12; package/engine.log and failed run.json.



Test conclusions and next actions

13 / QA runner shutdown

Conclusion: Initial source smoke leaks were traced to test-owned MP3 playback; later smoke and navigation shutdowns were clean.

Fix status: Runner stops audio/tweens, frees its scene and allows mixer drain time. Verified fixed for the harness.

To finish: Keep the cleanup regression. Do not transfer this root cause to the separate package leak without evidence.

Evidence: smoke/, verbose/, smoke-fixed/, smoke-confirm/, navigation/.

14 / Unseen-window attempt

Conclusion: The operator did not see the window and did not perform the earlier requested test.

Fix status: Evidence corrected; that run remains incomplete. A visible-window prerequisite is now in the QA workflow.

To finish: Do not use its 8,398-present trace as gameplay evidence. The separate visible retest supplies the later manual coverage.

Evidence: Launch report p. 15; package-manual/run.json; operator correction.

15 / Visible manual retest

Conclusion: Operator confirmed a usable window, five Play/Back cycles, match turns and no noticed issues. Seven declared checks pass.

Fix status: The failed capture workflow was retried successfully; shutdown was clean. Timing spikes remain separate findings.

To finish: Extend to full matches, save/reload and longer play. Do not label this short checklist a complete release sign-off.

Evidence: Launch report p. 16; package-visible/run.json and attestations.

16 / Reusable Swallowtail feature

Conclusion: Local CLI, GDScript extension, artifact retention, comparison, branded PDFs and skill are implemented and exercised.

Fix status: Initial QA feature is working; strict smoke comparison and validation passed. Product release remains separate.

To finish: Retain game-owned scenarios and evidence. Report conclusions must distinguish fixed, open and untested work; do not infer ship readiness from green checks.

Evidence: Upstream game-qa-baseline.md; QA validation receipts; source/package runs.



Is build 340 ready to ship?

HOLD final sign-off / release candidate

Build 340 is a credible release candidate, but the available evidence does not yet support unconditional PC ship sign-off. No critical player-facing crash, softlock or save-loss defect was reproduced. The hold is an advisory QA conclusion based on unfinished release coverage and unresolved performance/teardown findings, not proof that the game is broadly broken.

WHY THE CANDIDATE REMAINS PROMISING

840 regression checks passed on the export engine. The source scenario passed 46 checks. The visible package retest passed its seven checks with clean shutdown and no issue reported by the operator. The repeated source-menu hitch is fixed. These are strong results within their tested scope.

REQUIRED EVIDENCE BEFORE SIGN-OFF

1. Complete full matches in every advertised mode and the tutorial in the exact shipping package. Exercise win/loss results, restart, pause and return to title. Owner: game QA.
2. Verify ordinary progression and save/reload in the package: earned unlock, settings and progress survive exit/relaunch. Record the save fixture and outcome. Owner: game QA.
3. Run on the declared minimum-spec target, including the 62.875 MiB texture-cache tradeoff. Define supported hardware if it is not already declared. Owner: release owner and QA.
4. Close the warm-menu performance gate with ten timestamped package cycles. Triage the two >100 ms post-boot intervals, cold-start gap and first-entry hitch; fix reproducible interactive stalls or record an explicit, evidence-based risk acceptance. Owner: game developer.
5. Triage the headless shutdown leak. Supply a clean retest after a fix, or demonstrate headless-only scope and obtain a documented release decision. Owner: game developer.
6. Complete a focused 30-60 minute package soak and the advertised input/window checks, including controller support if offered, fullscreen and Alt-Tab. Owner: game QA.



What changes HOLD to READY

STOP-SHIP THRESHOLD

A reproducible crash, softlock, lost save or inability to complete an advertised core flow blocks release. None was demonstrated in the recorded runs. The remaining performance and headless-teardown findings need triage; their presence alone does not establish a critical player-facing defect.

CLOSURE AND FINAL DECISION

Use the build-340 hashes on page 12, or identify and retest a new candidate. Attach logs, operator results and action-marked frame traces for the required work.

Use the proposed ≤ 33.33 ms warm-menu peak on this machine as a retest target; set separate first-entry/startup and minimum-hardware budgets. These are proposed acceptance criteria, not previously approved requirements.

Move to READY only when required evidence is complete and remaining findings are fixed or explicitly accepted with scope, rationale, owner and follow-up. The release owner makes the final ship decision.

FOLLOW-UP AND RISK-ACCEPTANCE CANDIDATES

The repeated source menu hitch, regression-gate diagnostics, MCP key lookup and source-harness audio cleanup are closed in their measured scope; retain their regressions.

A cold-start delay confined to a responsive loading screen may be an accepted launch risk after measurement. It is not accepted automatically by this report.

Spacing audit findings and broader hardware/controller matrices can be scheduled after launch only if reviewed against advertised support and no core flow is affected.

Decision ownership

This is the QA recommendation for the recorded candidate, not an approval to publish. The release owner signs off the exact package and any accepted risks. No upload or release approval has occurred.